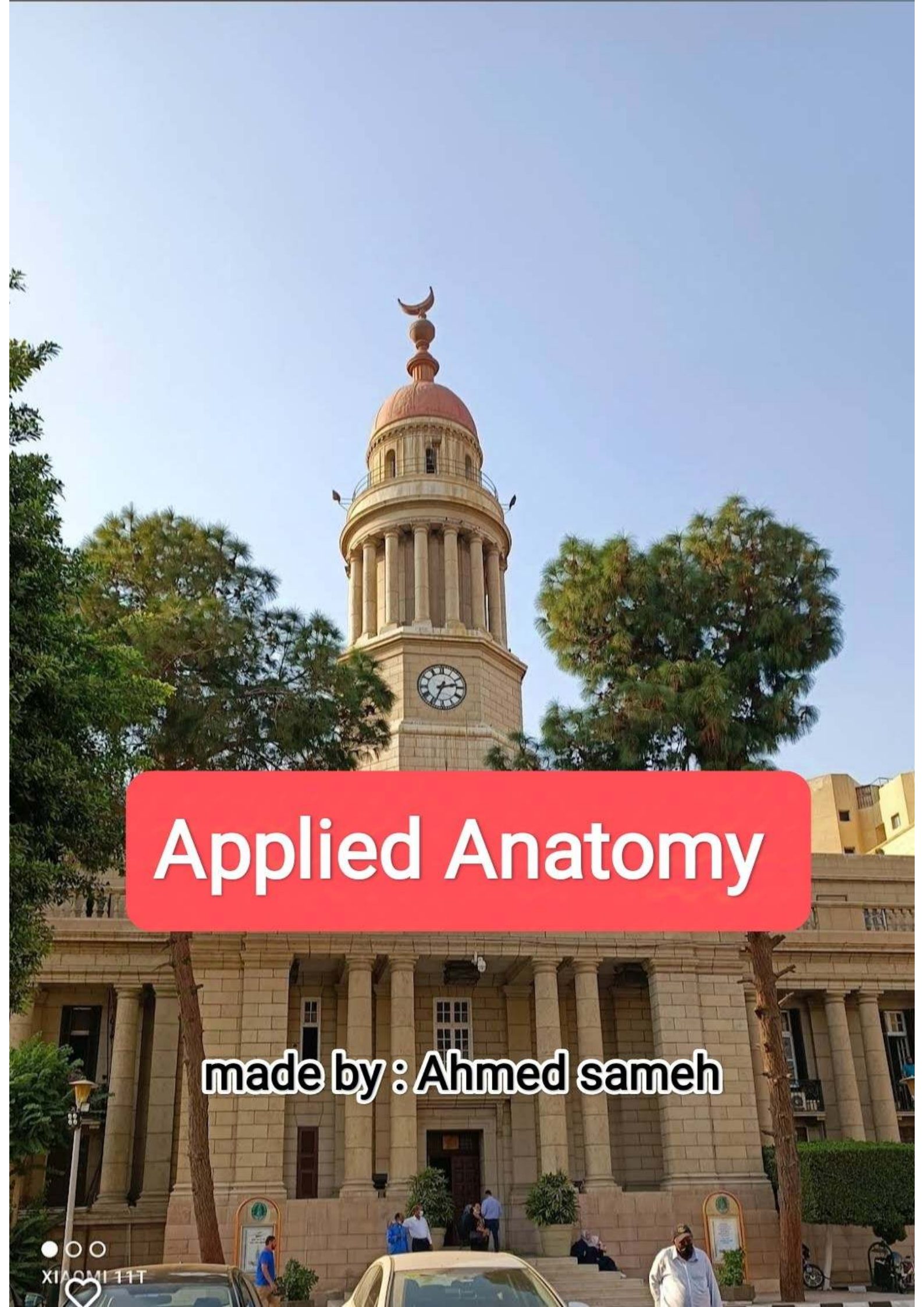




Applied Anatomy

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Lumbar puncture is procedure done by introducing a needle into the subarachnoid space either to obtain a sample of CSF or to inject a drug. It is safe to be done below of the level (L2) vertebra to avoid injury of the spinal cord.

6. Amount and pressure of CSF:

- The average amount of the CSF in adult is about 135 cc.
- It is constantly absorbed and produced in the rate of 400 to 500 cc daily, i.e. the CSF is replaced (renewed) three times daily.
- The pressure of the CSF is normally about 100 mm water. Any obstruction to the normal passage of the CSF especially at the narrow foramina and

- It is common that one of the arteries forming the circle of Willis may be very small or even completely absent without any clinical disturbances under normal conditions.
- Under normal conditions little mingling of the blood of the 2 sides occurs, but if one big artery becomes closed before it enters in the formation of the circle of Willis, the connections may open up.

N.B.: The cortical branches of the cerebral arteries anastomose with one another before, but not after, they enter the brain, where they become end-arteries.

The central branches are end-arteries.

Applied anatomy of the scalp:

- a-** Infection of connective tissue layer is usually limited due to the presence of the fibrous septa. On other hand cut wound in the scalp in this layer leads to severe bleeding due to its rich blood supply as well as the wall of the arteries are unable to contract due to its attachment to the surrounding fibrous septa.
- b-** The loose areolar tissue layer is the dangerous area of the scalp because infection can pass into the cranial cavity via the emissary veins present in it. Hemorrhage from these emissary veins is usually large and separates the scalp from the skull bones.

Applied anatomy:

I. Facial nerve injury (Facial palsy) is characterized by:

A- Paralysis of facial muscles leading to:

1. Dripling of saliva from the angle of the mouth in the affected side.
2. Accumulation of food in the vestibule of mouth in the affected side.
3. The patient can not close his eye in the affected side (the eyes remain open even during sleep).
4. Deviation of the angle of the mouth to the opposite side (healthy side).

B- Loss of taste sensation on the anterior 2/3 of tongue in the affected side.

II. Trigeminal neuralgia (Tic douloureux):

- It is a disease affecting the sensory root of the trigeminal nerve and caused by compression of the nerve by large blood vessel or by tumour.
- Trigeminal neuralgia is characterized by severe episodic pains restricted to the areas supplied by maxillary and/or mandibular nerves.

Applied anatomy:

Dangerous area of the face: Triangular area in the middle part of the face around the nose and upper lip. It is dangerous because infection in this area can spread and reach the cavernous sinus by this way: facial vein → deep facial vein → pterygoid venous plexus → emissary vein through foramen ovale → cavernous sinus.

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Applied Anatomy:

1- Inflammation of the parotid gland (parotitis): (bacterial or viral). The infection reaches the gland from the mouth through the duct of the gland or via the blood stream.

- The cardinal clinical picture of parotitis is severe pain and swelling of the parotid gland with raised or everted lobule of the auricle.
- The pain in the gland is due to its enlargement causing stretching of the tight parotid capsule.
- Parotitis causes pain during mastication due to close relation of the gland to the temporo-mandibular joint.

2- Parotid tumours usually arise from the part of the gland superficial to the branches of the facial nerve. Benign tumours do not affect the facial nerve but malignant ones compress and invade the nerve leading to facial nerve paralysis.

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Applied Anatomy:

The mandible is dislocated only forwards. With the mouth open, the condyles are present below the articular tubercle and sudden violence, even muscular spasm (convulsive yawn) may displace one or both condyles into the infratemporal fossa in front of the articular tubercle. Reduction of the condyles occurs by depressing the mandible posteriorly together with elevating the chin at the same time.

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Applied Anatomy:

The middle meningeal artery may be torn in temporal fractures or by injuries separating the dura mater from the bone, followed by haemorrhage between them (extradural haemorrhage). Trephine operation is necessary to reduce cerebral compression.

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Applied Anatomy:

Cavernous sinus thrombosis: It is a serious condition caused by transmission of infection from the dangerous area of the face to the cavernous sinus. It is characterized by fever and internal squint due to paralysis of abducent nerve. The eye becomes congested (black eye) and oedematous due to obstruction of its venous drainage and later on the eye is bulging to the outside (proptosis).

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Clinical importance:

Infection from outside the skull (in scalp and face) can spread to the dural venous sinuses inside the cranial cavity through these emissary veins leading to septic (infected) thrombosis of the sinuses e.g. cavernous sinus thrombosis.

Applied Anatomy:

Injury of the oculomotor nerve produces the following features:

- 1- Ptosis (drooping of the upper eyelid) due to paralysis of the levator palpebrae superioris.
- 2- Lateral squint, due to paralysis of the medial rectus and unopposed action of the lateral rectus.
- 3- Dilatation of the pupil due to paralysis of the sphincter pupillae.
- 4- Loss of accommodation due to paralysis of the ciliary muscle.

Applied anatomy:

The external jugular vein is adherent to the deep fascia at the point where it pierces it, if the vein is injured at this point it will not collapse and air embolism occurs.

Tributaries. External jugular vein is the only tributary.

Central venous catheter:

A tested technique of quickly accessing the major venous system. Its benefits include rapid administration of combinations of drugs, a route for nutritional support and fluid administration.

Central venous pressure is considered a direct measurement about heart activity. It is acquired by putting a central venous catheter into any of several large veins either in IJV or subclavian vein.

Applied Anatomy:

- 1- Enlargement of thyroid gland is called goiter. It may extend inside the thoracic cavity behind the sternum and is called retrosternal goiter causing dyspnea and venous congestion.
- 2- During thyroidectomy, manipulation of the superior thyroid artery may cause injury of the external laryngeal nerve. Manipulation of the inferior thyroid artery may cause injury of the recurrent laryngeal nerve.
- 3- Injury of the laryngeal nerves may lead to hoarseness of voice, aphonia or suffocation according to the degree of injury.
- 4- To avoid injury of the laryngeal nerves during thyroidectomy, the superior thyroid artery should be ligated as near as possible to the gland or even within the apex of the gland to avoid external laryngeal nerve injury. Ligation of the inferior thyroid artery should be as far as possible from the base of the gland to avoid injury of the recurrent laryngeal nerve.
- 5- During thyroidectomy we should leave the posterior part of the gland in which the parathyroid glands are embedded. Removal of the parathyroid glands during the surgery may lead to hypocalcemia and tetany.

Origin: Upper genial tubercle of mandible.

Insertion:

- a- **Upper and middle fibers** are inserted into the whole length of the ventral surface of the tongue.
- b- **Lower fibers** are inserted into the anterior surface of the upper part of the body of hyoid bone near the midline.

Nerve supply: Hypoglossal nerve (12th cranial nerve).

Action:

- 1- **One muscle:** Protrudes the tongue to the opposite side.
- 2- **Both muscles:** Protrude the tongue forwards and depress the central part of tongue. They also prevent the tongue from falling backwards during sleep as this may obstruct the air pathway.

In its nerve injury, tongue towards to the diseased side

Applied Anatomy:

Stone in the submandibular duct:

It is very common, 50 times more than that of the parotid duct because:

- 1- The secretion is thick and viscid.
- 2- The duct drains its secretion against gravity.
- 3- The duct lies on the floor of mouth where any food particle may enter it and form a nucleus for the stone.

The patient complains of swelling in the submandibular region which increases in size during meals and returns to its normal size in between meals.

Bell's palsy:

It is lower motor neurone lesion of the facial nerve which occurs spontaneously or without clear cause but recent MRI studies suggest that it may be caused by viral neuritis of the facial nerve at its beginning in the facial canal in the inner ear or in the adjacent brain stem.

It is characterized by:

- 1- Paralysis of the facial muscles of expression of the affected side leading to:
 - a. Dripping of saliva from the angle of the mouth in the affected side.
 - b. Accumulation of food in the vestibule of mouth in the affected side.
 - c. The patient can not close his eye in the affected side (the eyes remain open even during sleep).
 - d. Deviation of the angle of the mouth to the opposite side (healthy side).
- 2- Decrease lacrimation in the eye of the affected side which is controlled by the neurons of greater superficial petrosal nerve.
- 3- Hyperacusis (decreased tolerance of loud noise in the ear of the affected side due to paralysis of stapedius muscle).

In the vast majority of cases of Bell's palsy spontaneous full recovery of the case takes place after few weeks.

Applied anatomy:

Injury of glossopharyngeal nerve:

- 1- Absence of taste in the posterior 1/3 of tongue.
- 2- Absence of Gag reflex on the side of the lesion.

Applied Anatomy:

Injury of vagus nerve leads to:

- 1- Tachycardia and cardiac arrhythmia.
- 2- Vasovagal attack: reflex of involuntary nervous system due to stimulation of vagus nerve causes: bradycardia, hypotension and fainting attack.

Causes: some persons when they seen **blood** or subjected to any type of **fear**.

- 3- Injury of the pharyngeal branch leads to dysphagia and deviation of the soft palate (uvula) to the opposite side.
- 4- Injury of laryngeal nerves:

A- Injury of the external laryngeal nerve: The close relation of the external laryngeal nerve to the superior thyroid artery, the nerve may be cut when the artery is ligated and clamped during thyroid gland lobectomy.

Injury of the external laryngeal nerve leads to paralysis of the cricothyroid muscle and relaxation of the vocal cords these leads to:

1. The vocal cords become flaccid and flappy allowing excess leakage of air.
2. The voice become weak, low pitched and breathy voice accompanied by mild hoarseness. These effects occur in bilateral lesions, but in unilateral lesion these effects may not be noticeable.

B- Injury of recurrent laryngeal nerves (RLN):

- I. Unilateral RLN injury: It is common on the left side because it has long course in the thorax and the neck

Causes:

- a. In the thorax: RLN is damage by either cardiac surgery, trauma or bronchogenic carcinoma which is the commonest cause.
- b. In the neck: injury of Rt. or Lt. RLN in the neck can occur due to thyroid gland operation, cancer thyroid and penetrating neck trauma.

- II. Bilateral RLN injury: These injuries can occur during thyroidectomy or by cancer thyroid gland.

Clinical picture of the unilateral RLN injury

It leads to hoarse breathy voice, this hoarseness may be permanent or may become less severe with time as the opposite cord has the ability to hyperadduct and oppose the paralysed cord and thus close the glottis during phonation and coughing and these effects are enough to restore the full voice quality.

Clinical picture of the bilateral RLN injuries

- 1- **Partial bilateral RLN injuries:** induces paralysis of both abductors muscles of the vocal cords leading to adduction of the vocal cords leading to dyspnea and stridor. In this case tracheostomy is necessary for life.
- 2- **Complete bilateral RLN injuries:** induces paralysis of the abductor & adductor muscles of the vocal cords putting the vocal cords in position midway between abduction and adduction (cadaveric position) leading to aphonia = lose of speech.

Applied anatomy:**Injury of spinal part of accessory nerve:****Causes:**

Because it is subcutaneous nerve it can be injured by superficial cut wound or by lymph node biopsy.

The injured nerve leads to weakness and atrophy of sternomastoid and trapezius muscles leading to impairment rotatory movement of neck and face to the opposite side.

Palatine Tonsil

It is a mass of lymphatic tissue forming the lateral part of Waldeyer's ring.

Position:

It lies in the tonsillar fossa, between palatoglossal arch (in front) and palatopharyngeal arch (behind) in the lateral wall of the oropharynx.

Relations of palatine tonsil:

Above: Soft palate.

Below: Dorsum of pharyngeal part of tongue.

In front: Palatoglossal arch containing palatoglossus muscle.

Behind: Palatopharyngeal arch containing palatopharyngeus muscle.

Medially: Has a free superficial surface containing 12-15 orifices of tonsillar crypts which dip into the tonsil.

Laterally: The deep surface is covered by fibrous capsule and is related to the pharyngobasilar fascia and superior constrictor muscle forming together the tonsillar bed. Outside the superior constrictor muscle the palatine tonsil is related to the following blood vessels:

a- Ascending palatine artery.

b- Facial artery: giving its tonsillar branch.

c- Internal carotid artery lies 1 inch behind and lateral to the tonsil.

d- Paratonsillar vein.

Blood vessels of the tonsil:**Arterial supply:**

a- **Tonsillar branch of facial artery:** It is the main artery supplying the tonsil. It pierces superior constrictor muscle of pharynx to reach and supply the tonsil.

b- **Dorsal lingual arteries:** From the 2nd part of lingual artery.

c- **Ascending pharyngeal artery:** From the external carotid artery.

d- **Ascending palatine artery:** From facial artery.

e- **Lesser palatine arteries:** From the greater palatine artery.

Venous drainage: one or more veins leave the tonsil, traversing the superior constrictor muscle to join the paratonsillar, pharyngeal or facial veins.

Sensory nerves of the tonsil:

1- **Lesser palatine nerves:** From maxillary nerve (via sphenopalatine ganglion).

2- **Glossopharyngeal nerve:** Via its tonsillar (palatine) branch.

Lymph drainage of the tonsil:

Mainly into jugulo-digastric lymph node (one of the upper deep cervical lymph nodes)

Applied anatomy: inflammation of the tonsil is called tonsillitis and may act as septic focus which requires surgical removal of the tonsil called tonsillectomy. During the operation injury of paratonsillar vein may occur leading to haemorrhage. Referred pain from tonsil may radiate to the middle ear due to the common nerve supply of both (glossopharyngeal nerve 9th cranial nerve).

Applied Anatomy: (see page 107)

a- Injury to both external laryngeal nerves: Leads to paralysis of both cricothyroid muscles and relaxation of vocal cords, leading to weak, low pitched and mild hoarseness of voice.

b- Injury of recurrent laryngeal nerves:

1. On one side: It leads to hoarse breathy voice.
2. On both sides: complete injury leads to aphonia and incomplete injury leads to stridor.

c- Tracheostomy:

A tracheostomy is a procedure in which a hole is made in the trachea and a tube is inserted to enable ventilation.

In the emergency situation a tracheostomy is typically performed when there is obstruction to the larynx, either due to inhalation of a foreign body, severe edema secondary to anaphylactic reaction, or severe head and neck trauma.

In the emergency situation the cricothyroid ligament can be identified by simple palpation and a small needle can be inserted to establish an airway. The typical situation in which a tracheostomy is performed is in the calm atmosphere of an operating theatre. A small transverse incision is placed in the lower third of the neck anteriorly. The strap muscles are deviated laterally and the trachea can be easily visualized. Occasionally it is necessary to divide the isthmus of thyroid gland. An incision is made in the second and third tracheal rings and a small tracheostomy tube inserted.

After the tracheostomy has been in situ for the required length of time it is simply removed. The hole through which it was inserted almost inevitably closes without any intervention.

Applied anatomy:

Injury of the abducent nerve: Leads to medial squint due to paralysis of the lateral rectus.

Applied anatomy:

Injury of hypoglossal nerve characterized by:

- 1- Hemiatrophy of the affected side of the tongue.
- 2- On protrusion of the tongue, it deviates towards the affected side.

Applied Anatomy:

Pituitary tumours are characterized by 3 features:

- 1-** Clinical picture of endocrinal (hormonal) disturbance.
- 2-** Manifestations of increased intracranial tension as headache, vomiting and blurring of vision.
- 3-** Clinical picture due to pressure effects of the tumour, e.g. pressure of the pituitary tumour on the optic chiasma leading to a clinical condition called bitemporal hemianopia.